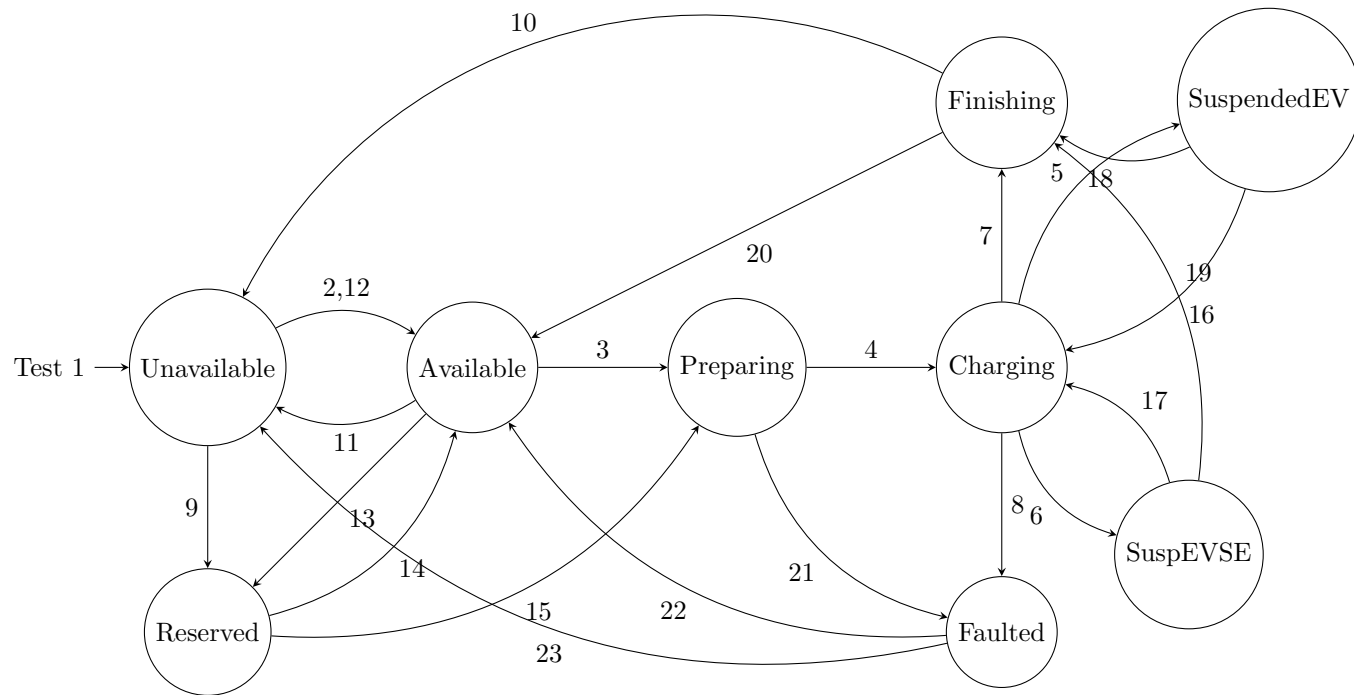


Evse.py Unit Test Report

MEA OCPP Certification (Automated)

January 14, 2026

State Transition Diagram



Unit Test Execution Results

ID	Test Case	State Check	Log Preview / Trace	Result
1	test_initial_state	Logic Check		PASSED
2	test_transition_to_available	Logic Check	Set the CP mode to EVSE Set the CP duty cycle to 100% Start the CP service Start SLAC in EVSE mode State transition: Unavailable -> Available	PASSED
3	test_transition_to_preparing	Logic Check	EV already connected State transition: Available -> Preparing	PASSED
4	test_transition_to_charging	Logic Check	"Energy transfer mode selected" received State transition: Preparing -> Charging Maximum voltage: 400 Maximum current: 32	PASSED
5	test_transition_to_suspended_ev	Logic Check	"DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 0A State transition: Charging -> SuspendedEV	PASSED
6	test_transition_to_suspended_evse	Logic Check	"DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 10A State transition: Charging -> SuspendedEVSE	PASSED
7	test_transition_to_finishing	Logic Check	"Session stopped" received State transition: Charging -> Finishing Closure type: 0	PASSED

ID	Test Case	State Check	Log Preview / Trace	Result
8	test_transition_to_faulted	Logic Check	"Session Error" received State transition: Charging -> Faulted Session error: 0: Unspecified	PASSED
9	test_transition_to_reserved	Logic Check	State transition: Unavailable -> Reserved Reservation set until 2026-01-01 12:00:00+00:00	PASSED
10	test_transition_to_unavailable_via_pending	Logic Check	Scheduling availability change to Inoperative State transition: Finishing -> Available Pending 'Inoperative' availability detected, transitioning to Unavailable. State transition: Available -> Unavailable	PASSED
11	test_transition_available_to_unavailable	Logic Check	State transition: Available -> Unavailable	PASSED
12	test_transition_unavailable_to_available	Logic Check	State transition: Unavailable -> Available ===== 23 passed in 0.09s =====	PASSED
13	test_transition_available_to_reserved	Logic Check	State transition: Available -> Reserved	PASSED
14	test_transition_reserved_to_available	Logic Check	State transition: Reserved -> Available	PASSED
15	test_transition_reserved_to_preparing	Logic Check	EV already connected State transition: Reserved -> Preparing	PASSED
16	test_transition_suspendedev_to_charging	Logic Check	"DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 10A State transition: SuspendedEV -> Charging	PASSED
17	test_transition_suspendedevse_to_charging	Logic Check	"DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 10A State transition: SuspendedEVSE -> Charging	PASSED
18	test_transition_suspendedev_to_finishing	Logic Check	State transition: SuspendedEV -> Finishing	PASSED

ID	Test Case	State Check	Log Preview / Trace	Result
19	test_transition_suspendedevse_to_finishing	Logic Check	State transition: SuspendedEVSE -> Finishing	PASSED
20	test_transition_finishing_to_available	Logic Check	State transition: Finishing -> Available	PASSED
21	test_transition_preparing_to_faulted	Logic Check	"Session Error" received State transition: Preparing -> Faulted Session error: 0: Unspecified	PASSED
22	test_transition_faulted_to_available	Logic Check	Resetting EVSE... State transition: Faulted -> Available	PASSED
23	test_transition_faulted_to_unavailable	Logic Check	State transition: Faulted -> Unavailable	PASSED

A Raw Unit Test Logs

```
1 ===== test session starts =====
2 platform darwin -- Python 3.12.3, pytest-9.0.2, pluggy-1.6.0 -- /Users/rus/development/MEA-V2G/venv/bin/python3.12
3 cachedir: .pytest_cache
4 rootdir: /Users/rus/development/MEA-V2G
5 configfile: pytest.ini
6 plugins: asyncio-1.3.0
7 asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=function, asyncio_default_test_loop_scope=function
8 collecting ... collected 23 items
9
10 tests/unit/test_evse_states.py::TestEvseStates::test_initial_state WHITE-beet-EI firmware version: 1.0.0
11 PASSED
12 tests/unit/test_evse_states.py::TestEvseStates::test_transition_available_to_reserved WHITE-beet-EI firmware version: 1.0.0
13 State transition: Available -> Reserved
14 PASSED
15 tests/unit/test_evse_states.py::TestEvseStates::test_transition_available_to_unavailable WHITE-beet-EI firmware version: 1.0.0
16 State transition: Available -> Unavailable
17 PASSED
18 tests/unit/test_evse_states.py::TestEvseStates::test_transition_faulted_to_available WHITE-beet-EI firmware version: 1.0.0
19 Resetting EVSE...
20 State transition: Faulted -> Available
21 PASSED
22 tests/unit/test_evse_states.py::TestEvseStates::test_transition_faulted_to_unavailable WHITE-beet-EI firmware version: 1.0.0
23 State transition: Faulted -> Unavailable
24 PASSED
25 tests/unit/test_evse_states.py::TestEvseStates::test_transition_finishing_to_available WHITE-beet-EI firmware version: 1.0.0
26 State transition: Finishing -> Available
27 PASSED
28 tests/unit/test_evse_states.py::TestEvseStates::test_transition_preparing_to_faulted WHITE-beet-EI firmware version: 1.0.0
29 "Session Error" received
30 State transition: Preparing -> Faulted
31 Session error: 0: Unspecified
32 PASSED
33 tests/unit/test_evse_states.py::TestEvseStates::test_transition_reserved_to_available WHITE-beet-EI firmware version: 1.0.0
34 State transition: Reserved -> Available
35 PASSED
36 tests/unit/test_evse_states.py::TestEvseStates::test_transition_reserved_to_preparing WHITE-beet-EI firmware version: 1.0.0
37 EV already connected
38 State transition: Reserved -> Preparing
39 PASSED
40 tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedev_to_charging WHITE-beet-EI firmware version: 1.0.0
41 "DC Charge Parameters Changed" received
```

```
42 EV maximum current: 32A
43 EV maximum voltage: 400V
44 EV ready: True
45 Error code: 0
46 SOC: 50%
47 EV target current: 10A
48 State transition: SuspendedEV -> Charging
49 PASSED
50 tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedev_to_finishing WHITE-beet-EI firmware version: 1.0.0
51 State transition: SuspendedEV -> Finishing
52 PASSED
53 tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedevse_to_charging WHITE-beet-EI firmware version: 1.0.0
54 "DC Charge Parameters Changed" received
55 EV maximum current: 32A
56 EV maximum voltage: 400V
57 EV ready: True
58 Error code: 0
59 SOC: 50%
60 EV target current: 10A
61 State transition: SuspendedEVSE -> Charging
62 PASSED
63 tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedevse_to_finishing WHITE-beet-EI firmware version: 1.0.0
64 State transition: SuspendedEVSE -> Finishing
65 PASSED
66 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_available WHITE-beet-EI firmware version: 1.0.0
67 Set the CP mode to EVSE
68 Set the CP duty cycle to 100%
69 Start the CP service
70 Start SLAC in EVSE mode
71 State transition: Unavailable -> Available
72 PASSED
73 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_charging WHITE-beet-EI firmware version: 1.0.0
74 "Energy transfer mode selected" received
75 State transition: Preparing -> Charging
76 Maximum voltage: 400
77 Maximum current: 32
78 PASSED
79 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_faulted WHITE-beet-EI firmware version: 1.0.0
80 "Session Error" received
81 State transition: Charging -> Faulted
82 Session error: 0: Unspecified
83 PASSED
84 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_finishing WHITE-beet-EI firmware version: 1.0.0
85 "Session stopped" received
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86 State transition: Charging -> Finishing
87 Closure type: 0
88 PASSED
89 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_preparing WHITE-beet-EI firmware version: 1.0.0
90 EV already connected
91 State transition: Available -> Preparing
92 PASSED
93 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_reserved WHITE-beet-EI firmware version: 1.0.0
94 State transition: Unavailable -> Reserved
95 Reservation set until 2026-01-01 12:00:00+00:00
96 PASSED
97 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_suspended_ev WHITE-beet-EI firmware version: 1.0.0
98 "DC Charge Parameters Changed" received
99 EV maximum current: 32A
100 EV maximum voltage: 400V
101 EV ready: True
102 Error code: 0
103 SOC: 50%
104 EV target current: 0A
105 State transition: Charging -> SuspendedEV
106 PASSED
107 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_suspended_evse WHITE-beet-EI firmware version: 1.0.0
108 "DC Charge Parameters Changed" received
109 EV maximum current: 32A
110 EV maximum voltage: 400V
111 EV ready: True
112 Error code: 0
113 SOC: 50%
114 EV target current: 10A
115 State transition: Charging -> SuspendedEVSE
116 PASSED
117 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_unavailable_via_pending WHITE-beet-EI firmware version: 1.0.0
118 Scheduling availability change to Inoperative
119 State transition: Finishing -> Available
120 Pending 'Inoperative' availability detected, transitioning to Unavailable.
121 State transition: Available -> Unavailable
122 PASSED
123 tests/unit/test_evse_states.py::TestEvseStates::test_transition_unavailable_to_available WHITE-beet-EI firmware version: 1.0.0
124 State transition: Unavailable -> Available
125 PASSED
126
127 ===== 23 passed in 0.09s =====
```